

Should we be excited or alarmed at the rate of progress of artificial intelligence?

This question requires students to be fully acquainted with both the aspirational goals and reality of the fields of robotics and Artificial Intelligence (AI). Students have to be able to make a persuasive case for whether Mankind should be **extremely excited** or **seriously alarmed** by the rapid progress of both these fields. Often times, students either possessed uneven knowledge of these two fields, or were unable to articulate compelling arguments for their respective promises or failures.

- This script is cogently and clearly argued with clear topic sentences.
- The student's knowledge of both robotics and AI is clearly exemplified through the use of relevant and contemporary examples.
- This essay also considers a number of alternative perspectives by pointing out that Man should not approach both these fields with unbridled optimism. While the student cautions against the loss of human abilities, she also points out insightfully that AI is nowhere near the deep learning systems of the human brain.

Just like the Luddites who worried about inventions during the Industrial Revolution superseding the need for their labour, particularly in the weaving industry, we, modern-day Luddites now fear ourselves being replaced by robots and artificial intelligence. Great strides have been made in such areas as machines have become more intricate and anthropoid, possibly reducing the need for human effort or collaboration. However, we should not be so alarmed by the rapid rise of robotics and artificial intelligence in the different facets of our everyday lives and instead be excited about their benefits akin to the increased efficiency and convenience brought about by the Industrial Revolution.

Machines can make our lives more comfortable. The development and commercialization of robotic helpers will greatly enhance the way we complete tasks. This runs the whole gamut from 'robo-nannies' taking care of our babies while we are at work to 'Robochef' which mimics the movements of eminent, top-tier chefs to prepare exquisite dishes in the comfort of our homes. Such machines can replace menial labour by performing tedious jobs that few are willing to undertake and alleviate problems associated with an ageing population prevalent in many developed countries. Conversely, they can also take on jobs that require a highly specialized yet learnable skill set, making such occupations and their associated benefits accessible to more people. Thus, machines and robots can greatly improve our lives as they are able to do our chores with ease and efficiency.

Another significant benefit of robotics and artificial intelligence is that they expand our computational capabilities. Humans are biological beings that are limited by innate restrictions. Neurons in our brain can only transmit signals at speeds about a hundred meters per second but machines circumvent this limitation as signals in transistors can travel at the speed of light, which is theoretically the maximum possible speed. The size of a machine or computer is not confined, unlike our brains, to a physical limit and can occupy enormous spaces such as warehouses. The processing speeds of machines are also much faster than our minds - easily a few gigahertz compared to our measly two hundred hertz. Thus, robotics and artificial intelligence can allow us to solve data-centric problems, optimize processes, and simulate appropriate models for further analysis. Thus, we should be excited as they augment our computational abilities and provide us with a tool that surpasses our inherent limitations.

Machine learning also enables us to solve complex problems which humans are unable to formulate. They are able to identify patterns when given large amounts of data and invent their own programme rules to solve abstract problems which we cannot quantify. An example would be identifying objects. Humans can recognize objects easily but machines face difficulties in doing so as there are no explicit rules governing the identity of an object. Machine learning enables us to find this rule and predict future cases that satisfy their criterion. For example, Facebook's DeepFace AI technology is able to correctly identify faces ninety-seven percent of the time, with frightening accuracy. Researchers have also utilized deep learning to predict the species of disease-carrying rodents. Thus, we should be optimistic about the development of robotics and artificial intelligence as they enable us to solve the increasingly abstract and multi-faceted problems facing our world.

However, we should not be overly enthusiastic about the benefits such technology can bring as it has certain drawbacks. Machine learning is still limited by the fact that machines require initial human judgment on a large test set of data in order to 'train' them. Machines are ultimately still unable to exercise autonomous thinking and can at best mimic it. Thus, we should not be overly optimistic about machine super intelligence and robots

passing the Turing's test as these predictions cannot be extrapolated from the current abilities of machines such as beating a grandmaster in chess.

Furthermore, artificial intelligence and robotics can lead to vast structural unemployment. As previously mentioned, robots can now replace our menial work and with the advent of developments such as virtual managers that can organize and govern a company with tremendous efficiency, our prospects may look bleak. However, we should not be daunted by this as such progress opens up job opportunities. New jobs are constantly being created which put skilled talents at the frontiers of such research, while computer science and information technology have widely popular fields. Thus, the loss of jobs due to developments in artificial intelligence is probably only transitory as our conventional jobs are displaced. It is important for us to work towards eventually obtaining new skills and knowledge, enabling the switch to different sectors of the industry just as workers during the Industrial Revolution did. Thus, we should not be so concerned about problems related to unemployment as it is only a transient displacement.

Besides displacing jobs, robotics and artificial intelligence can also threaten our fundamental right of privacy. Increased ease of surveillance due to developments in monitoring technology expedited by artificial intelligence has virtually eliminated our personal space. The Orwellian dystopia of a 'Big Brother' constantly watching Mankind now seems chillingly realistic. The National Security Agency (NSA) recently built a centre for surveillance which can store footage of more than a million people for a year which has terrifying repercussions for us. The development of commercial drones and miniscule cameras has increased the ease of spying. The confluence of such technology and the fact that drone regulations and anti-drone technology is still in its infancy poses a great threat to both our current and future definitions of privacy. The Edward Snowden incident revealed a shocking compromise between national security and individual privacy that has yet to be resolved, partly due to the dynamic nature of technological change which constantly surfaces new moral and legal issues. This issue could be further exacerbated by the development of robotics and artificial intelligence challenging on moral and social grounds our notions of privacy and its centrality to security issues.

Next, though unmanned air vehicles (UAV) or drones have seemingly revolutionized warfare by preventing the unnecessary loss of soldiers' lives, it has also raised a number of ethical problems, foremost of which includes the concern that drones do not understand the value of life and can lead to the callous trivialization of human life and the death of many innocent civilians in war. Ultimately, we could be desensitized to such vicarious killing if waging war is reduced to the operation of war drones which make killing as surreal and as easy as massacring entire armies during a video game. This raises attendant concerns about humans being more like machines while machines paradoxically become more anthropomorphic. Furthermore, the original intention of the US to develop drones, in an attempt to create a paradigm shift from nuclear warfare, may instead exacerbate the issue as China warns that such brandishing of technological superiority will increase the risk of nuclear warfare.

A lack of regulations governing the use of drones could pose dangerous problems regarding drone usage. A recent fire-fighting operation in California was grounded because recreational drones attached with cameras, were filming the incident. These drones obstructed and delayed aerial fire-fighting capabilities by up to 26 minutes, leading to severe damage of property. Such incidents reveal our trailing legislation and ethical guidelines which may eventually lead to our downfall if we do not tread precariously and cautiously on the road of robotics and artificial intelligence.

Ultimately, I believe people have come to be intelligent and discerning enough to recognize the dangers of such technology and our lack of wisdom in controlling it. Thus, I am confident that people can learn from such incidents and develop our ethical guidelines and legislation in tandem with technological progress before robotics and artificial intelligence envelopes our lives completely. The dystopian prospect of super-intelligence replacing mankind still, however, looks unlikely as machines are still unable to exercise independent thinking. The 'grey goo' scenario where self-replicating nanobots take over the world also seems unlikely with our current technology. However, even if such cases were probable, I believe that we, 'chimpanzees', can utilize our limited intelligence to teach such new advanced life-forms or hyper-intelligence our values and ethical systems. We should be extremely excited by the rapid progress being made in the fields of robotics and artificial intelligence. Yet, at the same time, we must not allow this to turn into unbridled optimism and we need to tread cautiously and improve ourselves such that we will be enlightened and wise enough to handle such advanced technology when it becomes a reality.

